Shubhangi S. R. Garnaik

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Research Interests

Machine learning systems, augmented and virtual reality, computer vision, secure mobile systems, video analytics, AI for sustainability, and human-computer interaction.

Education

Ph.D., Computer Science 2024-Pursuing
Stony Brook University
Advisor: Prof. Jihoon Ryoo

M.S., Computer Science 2022–2023
Stony Brook University. GPA: 3.92/4.0

Bachelor of Engineering, Computer Engineering 2015–2019
Pune University. GPA: 3.33/4.0

Research Experience

Graduate Researcher in AI2S Lab 2024–Present
Stony Brook University / SUNY Korea

- Conducting research on AI-enabled augmented reality, streaming-system workflows, payment security with a focus on deployable applied-AI prototypes.
- Developing multimodal AI research for audio-visual scene understanding, real-time perception, and interactive intelligent systems.

Research Intern in MEIC Lab Jul. 2025–Dec. 2025
SUNY Korea, Stony Brook University

- Contributed to computer-vision system implementation and reproducible deployment workflows for applied-AI research prototypes.
- Designed edge-case tests for backend API endpoints, validated system behavior, and documented failure modes to improve research-software reliability.
- Implemented and containerized a 3D Gaussian Splatting pipeline spanning preprocessing, training, postprocessing, and rendering; prepared run scripts and documentation.

Research Assistant, School-Company Collaborative Project 2025
SUNY Korea / Taeyang Metal Industrial Co., Ltd.

- Developed a computer vision framework for real-time press/forging monitoring by processing camera footage of analog pressure gauges and digital graph screens.
- Audited large-scale production recordings to document critical image-quality constraints, mitigating issues like high-frequency camera vibration, inverted spatial orientations, and low-frame-rate signal degradation.
- Engineered anomaly-analysis workflows for industrial telemetry videos, identifying micro-level signal deviations and graph anomalies to predict structural fractures and equipment failures.

Professional Experience

Data Scientist Mar. 2024–Oct. 2024
Thanks Carbon, Korea

- Optimized alternate wetting and drying techniques with deep learning models to support reduced carbon emissions in rice agriculture.
- Developed predictive models using weather and rice-yield data for targeted regions.
- Applied deep learning techniques for predictive analytics in agricultural efficiency and environmental sustainability.

Software Test Engineer / Associate Technical Sales Engineer Jun. 2019–Sep. 2021
Zensoft / Qualitia, India

- Automated client software testing, reducing testing time and improving software quality through structured test workflows.

Awards and Scholarships

- Winner, Decompiler HACKATHON THE X 2026, SUNY Korea May 2026
- Idea Finalist, Flipkart Hackathon 2019
- Conference Scholar, Distributed Systems Conference, Pune 2019

Publications

1. **Shubhangi S. R. Garnaik**, A. Balasubramanian, N. Balasubramanian, and J. Ryoo. "ArtSpeak: An Interactive AR Application for Lifelike Speaking with Art Portraits." *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2025.
2. Hoyoung Kim, Azimbek Khudoyberdiev, **Shubhangi S. R. Garnaik**, Arani Bhattacharya, and Jihoon Ryoo. "CLOUD-CODEC: A New Way of Storing Traffic Camera Footage at Scale." *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)*, Aug. 2025.
3. Azimbek Khudoyberdiev, **Shubhangi S. R. Garnaik**, Ha Young Kim, and J. Ryoo. "A Study of the Man and Unmanned Teaming System for Improving Efficiency in a Fulfillment Center." *IEEE Access*, 2023.
4. **Shubhangi S. R. Garnaik**, D. Lee, Y. Kim, and J. Ryoo. "MINSU: Precision Quantity Counter with DNN-based Volume Estimation." *International Conference on ICT Convergence (ICTC)*, 2023.
5. **Shubhangi S. R. Garnaik**, Y. Kim, and J. Ryoo. "SQR: Secure QR Transaction with Randomized Rotation." *International Conference on ICT Convergence (ICTC)*, 2022.

Posters and Workshop Papers

Shubhangi S. R. Garnaik and Jihoon Ryoo. "Enhancing Liver Disease Prediction using Machine Learning Models: A Comparative Study of Oversampled and Synthetic Data Approaches." Presented at KICS, 2023.

Shubhangi S. R. Garnaik and Jihoon Ryoo. "Exploring Sentiment Analysis of Silicon Valley Bank Collapse: Deep Learning Approaches on Google News Headlines." Presented at KICS, 2023.

Teaching Experience

Teaching Assistant — Aug. 2025–Spring 2026
SUNY Korea / Stony Brook University

- CSE 113: Foundations of Computer Science, Spring 2026
  Instructor: Prof. Zhoulai Fu
- CSE 310: Computer Networks, Fall 2025
  Instructor: Prof. Sehwan Yoo
- CSE 373: Analysis of Algorithms, Fall 2025
  Instructor: Prof. James Finn

Technical Skills

Programming: Python, Java, C#, HTML/CSS.
Developer Tools: Git/GitHub, Linux, Jupyter Notebook, Google Colab, VS Code, Eclipse, Unity, Android Studio, Docker.
AI/ML & Computer Vision: PyTorch, TensorFlow, OpenCV, deep learning, computer vision, video analytics, 3D Gaussian Splatting workflows, Snntorch, Lava.
Testing & Documentation: Backend API testing, software test automation, model documentation, technical presentation.

Activities & Services

- Orientation Leader at SUNY Korea, supporting freshman student transition.
- Organized an inter-college coding competition in Pune to support logical thinking and coding proficiency.

References

Jihoon Ryoo (jihoon.ryoo@stonybrook.edu)
Assistant Professor, Department of Computer Science, Stony Brook University / SUNY Korea

Aruna Balasubramanian (arunab@cs.stonybrook.edu)
Associate Professor, Department of Computer Science, Stony Brook University